Cross-Platform Prompting:
Evaluating Diverse Techniques in AI-Powered Text Summarization

Course: Prompt Engineering & AI Applications | Level: Undergraduate

Task Domain: Text Summarization | Article: The Basics of Blockchain Technology

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AIM

To evaluate and compare the effectiveness of prompting techniques — Zero-Shot, Few-Shot, Chain-of-Thought, and Role-Based — across AI platforms (ChatGPT, Gemini, Claude, Copilot) for the specific task of summarizing a 500-word technical article on Blockchain Technology.

SCENARIO

You are part of a content curation team for an educational platform that delivers quick summaries of research papers to undergraduate students. Your task is to summarize a 500-word technical article on 'The Basics of Blockchain Technology' using multiple AI platforms and prompting strategies, then determine which combination of Prompting Technique + Platform provides the best summary.

EVALUATION CRITERIA

1.	**Accuracy**	Factual correctness of the summary
2.	**Coherence**	Logical flow and readability
3.	**Simplicity**	Ease of understanding for students
4.	**Speed**	Response generation time
5.	**User Experience**	Overall ease of platform use

PROMPTING TECHNIQUES USED

Zero-Shot Prompting

Description: The AI is given no examples; it relies purely on its pre-trained knowledge.

> ***Sample Prompt:*** *Summarize the following article on Blockchain Technology in 150 words for undergraduate students.*

Outcome: Fast, minimal setup, good baseline. May lack focus or depth.

Few-Shot Prompting

Description: Two or three example summaries are provided before the actual request to guide output style.

> ***Sample Prompt:*** *Here are two example summaries of technical articles: [Example 1]... [Example 2]... Now summarize this blockchain article similarly.*

Outcome: Higher consistency and style alignment. Requires crafting good examples.

Chain-of-Thought (CoT) Prompting

Description: The model is instructed to reason step-by-step before generating the final summary.

Sample Prompt: *Step 1: Identify the main topic. Step 2: List key concepts. Step 3: Organize logically. Step 4: Write a student-friendly summary.*

Outcome: More structured and accurate output. Slightly longer response time.

Role-Based Prompting

Description: The AI is assigned a persona — e.g., an expert educator — to shape its tone and depth.

Sample Prompt: *You are an expert educational content summarizer. Summarize the following blockchain article for undergraduate students in 150-200 words.*

Outcome: Tone-appropriate, engaging summaries. Best for educational contexts.

PLATFORM-WISE SUMMARY OUTPUTS

The same article was submitted to four platforms using each prompting technique. Below are representative summaries generated by each platform using Role-Based Prompting (the highest-performing technique).

ChatGPT (GPT-4o)

Blockchain is a decentralized digital ledger that records transactions across multiple computers. Each record, called a 'block,' is linked to the previous one via a cryptographic hash, forming an immutable 'chain.' Because no single entity controls it, the system is resistant to tampering. Transparency is built-in — all participants can view the same data simultaneously. Blockchain is used in cryptocurrencies (Bitcoin, Ethereum), supply chain tracking, digital voting, healthcare data sharing, and smart contracts. Its combination of security, transparency, and decentralization makes it a transformative technology for the digital era.

Google Gemini

Blockchain technology functions as a distributed ledger system where data is organized into blocks and secured using cryptographic hashes. Each block connects to its predecessor, making unauthorized modification practically impossible. The decentralized nature means no single server or authority manages the data. This peer-to-peer structure ensures high transparency and collective verification. Real-world applications span financial services, logistics, medical records, and governance. Gemini's summary was notably concise and student-friendly, avoiding heavy technical jargon.

Claude (Anthropic)

Blockchain is a digital record-keeping system that stores data in linked blocks across a decentralized network. Each block contains a unique hash — a digital fingerprint — that includes the previous block's hash, making the chain tamper-evident. No central authority controls it; instead, thousands of computers (nodes) validate transactions. This ensures transparency: everyone sees the same ledger. Applications include Bitcoin, supply chain management, healthcare records, e-voting, and smart contracts. Claude's output balanced technical accuracy with accessibility, making it ideal for undergraduate comprehension.

Microsoft Copilot

Blockchain is essentially a shared digital notebook distributed across thousands of computers. Each entry (block) is locked with a unique code (hash) and linked to the previous one, preventing tampering. There is no central authority — changes require agreement from the majority of the network. This makes it highly secure and transparent. Industries from finance to healthcare are adopting blockchain for its ability to create trust without intermediaries. Copilot's output used approachable analogies that resonated well with student audiences.

TABLE 1: Platform × Technique Evaluation Matrix

Scores are out of 5. Ratings based on evaluated outputs across all four platforms and techniques. Higher score = better performance.

Platform	Technique	Accuracy (/5)	Coherence (/5)	Simplicity (/5)	Speed (/5)	User Exp (/5)	Total (/25)
ChatGPT	Zero-Shot	3.5	3.5	3.5	4.5	4.0	19.0
ChatGPT	Few-Shot	4.0	4.0	4.0	4.0	4.0	20.0
ChatGPT	CoT	4.5	4.5	4.0	3.5	4.0	20.5
ChatGPT	Role-Based	4.5	4.5	4.5	4.0	4.5	22.0
Gemini	Zero-Shot	3.5	3.5	4.0	4.5	4.0	19.5
Gemini	Few-Shot	4.0	4.0	4.0	4.0	4.0	20.0
Gemini	CoT	4.0	4.0	4.5	3.5	4.0	20.0
Gemini	Role-Based	4.5	4.5	4.5	4.5	4.5	22.5
Claude	Zero-Shot	4.0	4.0	4.0	4.5	4.5	21.0
Claude	Few-Shot	4.5	4.5	4.5	4.0	4.5	22.0
Claude	CoT	4.5	5.0	4.5	4.0	4.5	22.5
Claude	Role-Based	5.0	5.0	5.0	4.5	5.0	24.5
Copilot	Zero-Shot	3.5	3.5	4.0	4.5	3.5	19.0
Copilot	Few-Shot	4.0	3.5	4.0	4.0	3.5	19.0
Copilot	CoT	4.0	4.0	4.0	3.5	3.5	19.0
Copilot	Role-Based	4.5	4.0	4.5	4.0	4.0	21.0

TABLE 2: Best Prompting Technique per Platform

Platform	Best Technique	Total Score (/25)	Strengths
ChatGPT	Role-Based	22.0/25	Balanced accuracy & UX; clear language
Gemini	Role-Based	22.5/25	Excellent simplicity & fast responses
Claude	Role-Based	24.5/25	Highest accuracy, coherence & UX
Copilot	Role-Based	21.0/25	Good analogies; moderate coherence

TABLE 3: Average Scores by Prompting Technique

Prompting Technique	Avg Accuracy	Avg Coherence	Avg Simplicity	Avg Speed	Avg UX	Avg Total
Zero-Shot	3.63	3.63	3.88	4.50	4.00	19.38
Few-Shot	4.13	4.00	4.13	4.00	4.00	20.25
CoT	4.25	4.38	4.25	3.63	4.00	20.50

Role-Based	4.63	4.50	4.63	4.25	4.50	22.50

ANALYSIS & OBSERVATIONS

- ➤ **Zero-Shot prompting** was the fastest but produced generic summaries with less focus on student-friendly language. Suitable only for quick drafts.
- ➤ **Few-Shot prompting** improved consistency and style, but required quality examples. Beneficial when a specific output format is needed.
- ➤ **Chain-of-Thought prompting** produced the most structured outputs with clear logical flow, but was slightly slower due to step-by-step reasoning.
- ➤ **Role-Based prompting** consistently outperformed others across all five criteria — especially in accuracy, simplicity, and user experience. It is highly recommended for educational content generation.
- ➤ **Claude performed best overall**, particularly with Role-Based prompting (24.5/25), delivering technically accurate and pedagogically sound summaries.
- ➤ **Gemini excelled in speed and simplicity**, making it a strong choice when turnaround time is critical.

RESULT

The experiment successfully evaluated four prompting techniques across four AI platforms for the task of summarizing a technical article on Blockchain Technology. The **Role-Based Prompting** technique combined with the **Claude platform** yielded the best overall performance with a score of **24.5 out of 25**, excelling in accuracy, coherence, simplicity, and user experience. Among techniques, Role-Based Prompting averaged **22.50/25** across all platforms — significantly outperforming Zero-Shot (19.38), Few-Shot (20.25), and CoT (20.50). This experiment concludes that for educational text summarization tasks, **Role-Based Prompting on Claude is the most effective combination**, while Gemini with Role-Based prompting is the best alternative when speed is a priority.